Module-2

1. Explain matplotlib API, with suitable Examples.

Matplotlib API:

Matplotlib is a popular python library used creating graphs, charts & data visualization It is widely used in data science.

the matplotlib API Provides functions & classes.

- Line plots
- Bar charts
- Histograms
- Scatter plots
- Pie charts
- Bar plots

Function	Purpose
Plt.Plot()	create a line plot
Plt.bar()	create a bar chart
Plt.scatter()	create a scatter Plot
Plt.hist()	create a histogram
Plt.pie()	create a pie chart

Basic structure of matplotlib:

1. import matplotlib
2. Prepare the data
3. Select the type of plot
4. Add filters & labels
5. Display the plot

Advantages of matplotlib:

1. Easy to use
2. many Plot types
3. Highly customization
4. works with numpy & pandas.

2. Different line plot, Bar plot, Histograms Scatter plot?

Different types of plot are used for different purpose in data visualization.

→ line plots
→ Bar plots
→ Histograms
→ Scatter plots
→ four commonly used plots.

1) line plots: A plot represents data points connected by lines. Mainly used to show trends or changes. an ordered sequence.

2) Barplot: A bar plot represents data using rectangular bars. The height or length of each bar represents the value of a category.

3) Histogram: A histogram is used to represent the distribution of numerical data. It divides data into intervals called bins & shows how many observations fall into each interval.

4) Scatter plot: A scatter plot display individual data points using dots. It is mainly used to examine the relationship or correlation between two numerical variables.

2. Explain seaborn facet grid & categorical plots seaborn.

- Seaborn is a python data visualization lib built on top of matplotlib. It provides a high-level interface for creating attractive & informative.

→ It is commonly imported as
import seaborn as sns
Two imp concepts in seaborn are
1. face grid
2. categorical plots.

seaborn facegrid:

Facegrid is a seaborn features used to create multiple plots based on different subjects of a data set.

Syntax:

sns.facegrid(data, col="column")

Here data = dataset

col = variable used to create columns.

Example:

Suppose we have a dataset containing

→ Student

→ Gender

→ Marks

→ Department.